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"Shuttle" in Polysulfide Shuttle: Friend or Foe?

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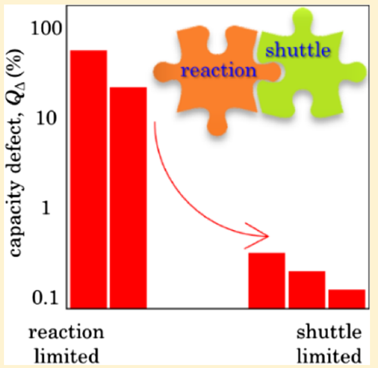
Supporting Information

ABSTRACT: The polysulfide shuttle effect, where sulfur species reach the negative electrode surface and undergo chemical reduction, is believed to be a major bottleneck in lithium−sulfur chemistry. The importance of this phenomenon is often judged based on phenomenological arguments, which do not account for mesoscale complexations. This work presents a comprehensive investigation of the coupled interactions arising from speciation, concentrated electrolyte solution, and reaction time scales. Polysulfide transport consists of diffusion and migration, which determines the net flux. This study demonstrates that the polysulfide shuttle effect can be bounded between reaction-limited and shuttle-limited regimes, depending on the operational extremes. At high sulfur to electrolyte ratio (i.e. lean electrolyte condition), the polysulfide shuttle effect may not necessarily attribute to the cell performance limitation, believed contrarily otherwise.



INTRODUCTION

Lithium−sulfur (LiS) chemistry has received a considerable attention of late given its superior theoretical promise as compared to the state-of-the-art energy storage solutions.[1−5] It is a uniquely interesting electrochemical system where electroactive couples are different at the anode (Li|Li^+) and cathode (S_8|S_x^{2-}, $x \in [1, 8]$). Sulfur species transfer to the anode and chemically react with Li, thus leading to a reduced electrochemical capacity (in contrast, the Li^+ reaching the cathode is already in the highest oxidation state and can only partake in nonredox interactions of its own, such as Li_2S precipitation). This phenomenon has been termed as the polysulfide shuttle effect and is traditionally believed to be the source of irreversibilities associated with the LiS system.[6−15] Surprisingly, even if the polysulfide shuttle is one of the most often discussed facets of LiS, a fundamental understanding is lacking as the detailed mesoscale interactions have never been probed and explicit investigations have either measured cell-level observables[16−24] or focused on the anode surface.[25−27] Recent studies have identified the presence of multimodal physicochemical interactions fostering nonideal response,[28] such as precipitation−dissolution dynamics,[29−31] interfacial kinetics,[32−35] concentrated solution effects,[36−39] and poromechanical progression[40,41] to name a few. The nature of their convolution with the polysulfide shuttle remains unknown.

The polysulfide shuttle is essentially composed of two distinct events: shuttling of (electrolyte phase) sulfur species and their subsequent chemical reduction at the anode. The species flux (i.e., shuttle) is strongly related to the cathode evolution[29] and electrolyte speciation.[39] The present study is aimed at delineating the intricate correlations among the polysulfide shuttle and other dynamical progressions present therein. Contributions from responsible chemical reactions vary as sulfur is reduced further. During the rest state, the chemical redox reaction is responsible for self-discharge. The capacity defect defines the strength of chemical redox, when electrochemical reactions are active (i.e., charge/discharge). On the other hand, difference potential characterizes the severity of these anode-centric reactions during the rest phase (i.e., self-discharge). The nonmonotonic trends resulting from differences in timescales for chemical and electrochemical reactions are analyzed as well. Sulfur and electrolyte contents affect the polysulfide shuttle as they affect cathode starvation modes and electrolyte transport dynamics.

PHYSICOCHEMICAL DESCRIPTION OF THE SHUTTLE EFFECT

Figure 1a presents the potential evolution when sulfur impregnated in a mesoporous carbon structure is reduced at 1 C current [the corresponding current value is linearly proportional to volumetric sulfur loading, refer to expressions (S27) and (S28)]. The cell is composed of a porous carbon cathode with 10% vol solid sulfur and 70% initial porosity. Relevant modeling details are supplied in Supporting Information. Different forms of sulfur are generated at the cathode in response to physical changes (e.g., dissolution of solid sulfur leading to dissolved sulfur in the electrolyte) and electrochemical reactions (i.e., reduction of sulfur into successively smaller chain polysulfides) and can reach the anode. The incoming flux of these electrolyte species is composed of two distinct transport modes (i) diffusion—related to concentration gradients and (ii) migration—due to the electrolyte phase potential gradient (Figure 1d). These sulfur species interact with the Li metal anode via chemical

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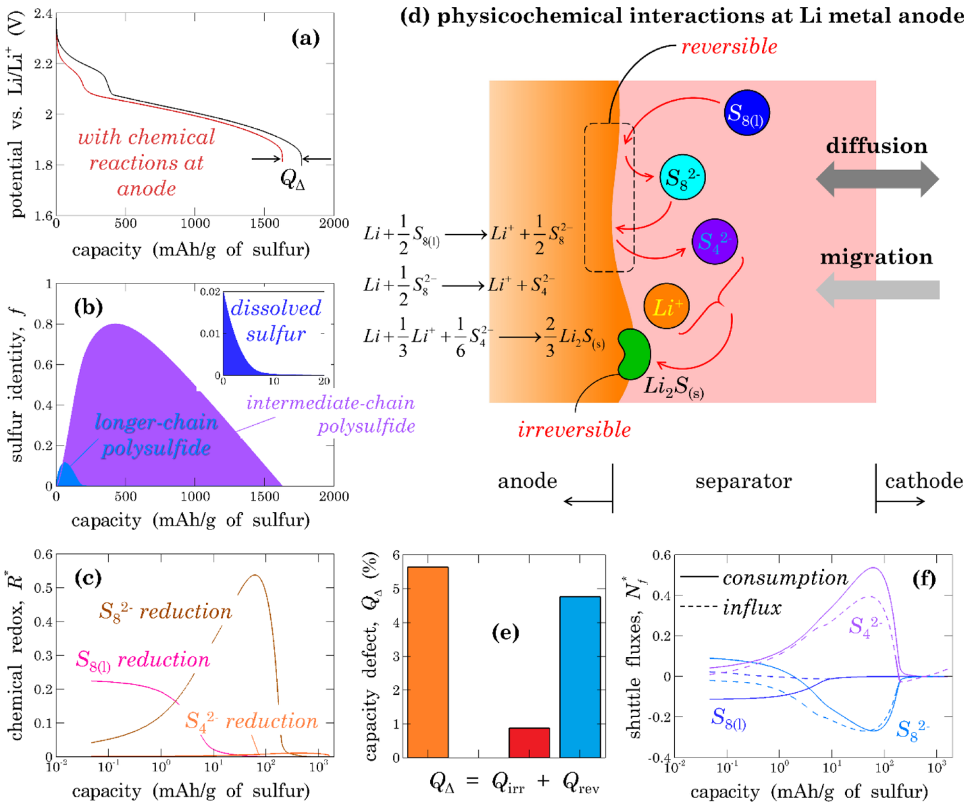

Figure 1. Physicochemical interpretation of the polysulfide shuttle effect: (a) effect of chemical redox reactions at the anode surface on potential evolution upon sulfur reduction, (b) chemical transitions of sulfur coincide with (c) predominant redox at the anode surface, (d) a schematic illustrating the species influx from the cathode and subsequent consumption (because of chemical reactions) at the anode surface, (e) this chemical reduction of sulfur molecules reduces the achievable capacity that has both reversible and irreversible defects, and (f) comparison of influx and consumption of sulfur species reveal that not all that comes in from the cathode get reacted at the anode and contribute to the capacity defect.

Table 1. Terminology Associated with the Physicochemical Description of Polysulfide Shuttle Effect

terms	description
capacity defect, Q_Δ	is capacity loss due to chemical reduction at the anode surface; there are other factors affecting capacity, for example, surface passivation
reversible capacity defect, Q_{rev}	is a portion of the capacity defect caused by chemical reduction where the reduced species returns back to the electrolyte
irreversible capacity defect, Q_{irr}	is a portion of the capacity defect caused by chemical reduction where the reduced species deposits at the anode
voltage defect, Φ_Δ	is the difference between the rest phase voltage and open-circuit potential; it is nonzero for a self-discharging cell; if a cell is let to self-discharge, all sulfur is reduced via chemical reactions at the anode and $Q_\Delta \rightarrow 100\%$
chemical overpotential, η_{ch}	the reaction rate, k, for chemical reduction at the anode is correlated to exchange current density, i_0, for the corresponding reaction at the cathode; $k = \frac{i_0}{F}\exp\left(-\frac{F\eta_{ch}}{2RT}\right)$ chemical overpotential, η_{ch}, characterizes the reactivity of the anode surface
influx	the flux of polysulfide species from the cathode to the separator
consumption flux	the rate of polysulfide chemical reduction at the anode−separator interface

redox reactions, namely (S5−S7) (Li loses an electron and oxidizes, whereas sulfur species accept electrons and get reduced further; no electron is exchanged with the external circuit, hence such reduction is purely chemical in nature). Note that these reactions are different than the electrochemical reduction of sulfur at the cathode where electrons are exchanged with the external circuit. Figure 1a depicts the effect of sulfur reduction at the anode on potential evolution. Quantitatively, it manifests as a capacity difference, Q_Δ. The capacity defect, Q_Δ, is solely the contribution of anode surface reactions (refer to Table 1 for terminology). As authors have recognized earlier,[29,39] there are other possible sources of capacity loss such as surface passivation or transport losses, and all these combined together show a capacity less than 1675.12 mAh/g of solid sulfur.

Let us deconvolve these chemical interactions at the anode surface. Similar to the electrochemical reduction events at the cathode, a commensurate set of reactions takes place at the anode surface and exhibits chemical nature (Figure 1d). Figure 1b sketches the evolution of different sulfur forms in the cell (refer to Section S3: Sulfur Balance Sheet for specific queries). For the particular rate of sulfur reduction being discussed here (Figure 1; 1 C), the life span of various sulfur species overlap. Given the finite solubility of dissolved sulfur, $S_{8(l)}$, it is never the dominant form of sulfur and coexists with other polysulfides. Longer-chain polysulfide, S_8^{2-}, achieves its highest concentration when the dissolved sulfur is nearly consumed

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(compare Figure 1b and inset). Once the potential drops below that for reduction of longer-chain polysulfide to the intermediate-chain, these medium-chain polysulfides start forming. Beyond this point, one observes a continuous dissolution of solid sulfur and its reduction to longer-chain and subsequently intermediate-chain polysulfides. The occurrence of these different sulfur species in the electrolyte (Figure 1b) with appropriate chemical reduction rates at the anode surface was compared (Figure 1c). These reaction rates have been normalized using the applied current density for a consistent interpretation across different operating conditions (note that each of these chemical reactions (S5−S7) are written as single-electron exchange processes)

$$R^* = \frac{FR}{J_{\text{app}}} \tag{1}$$

Figure 1b,c jointly reveals that the dominant chemical redox reaction closely follows the identity of sulfur species in the electrolyte phase (Figure 1d). The rate of each of these reactions depends on two factors: reactivity and reactant concentration. Reactivity is characterized by the reaction rate constant (because these reactions are analogous to those taking place at the cathode, one would expect a qualitatively similar order of reactivity). This explains the difference in average reaction rates in Figure 1c. Temporal variation in local reactant concentration correlates to the reaction rate evolutions for respective reduction events in Figure 1c. Despite the fact that these reactions are all chemical in nature, their contributions to the capacity defect, Q_Δ, are qualitatively different. Reduction of dissolved sulfur (reaction S5) and longer-chain polysulfide (reaction S6) releases species in the electrolyte phase which in principle can be oxidized back to recover the capacity lost in the respective chemical reactions. On the other hand, the chemical reaction of intermediate-chain polysulfide (reaction S7) deposits Li_2S at the anode and cannot be oxidized electrochemically. Thus, the total capacity defect (because of chemical redox at the anode) has two types of contributions: reversible, Q_{rev}, and irreversible, Q_{irr}.

$$Q_\Delta = Q_{\text{rev}} + Q_{\text{irr}} \tag{2}$$

$$Q_{\text{rev}} = \int_{t_{\text{operation}}} F(R_5 + R_6)\text{d}t \tag{3}$$

$$Q_{\text{irr}} = \int_{t_{\text{operation}}} FR_7 \text{ d}t \tag{4}$$

Figure 1e inspects the contributions from these different forms for 1 C operation. The values are normalized using the theoretical maximum capacity for a better appreciation.

These chemical reactions are sustained based on the availability of sulfur species in the vicinity of the anode, which are primarily replenished because of the incoming flux from the cathode. Because no electrochemical reactions take place in the separator, the ionic current at any cross section along the thickness direction is constant and equal to the applied current density

$$I = F(N_{\text{p}} + z_{\text{a}}N_{\text{a}} - 2N_{\text{x}} - 2N_{\text{y}}) = J_{\text{app}} \tag{5}$$

Going from the anode to cathode, the net ionic current remains invariant, but the contribution from individual species varies based on local concentration distributions, which are not necessarily monotonic. Take longer-chain polysulfide for an instance: at first, electrochemical reduction generates a positive slope in S_8^{2-} concentration at the cathode−separator interface which amounts to an influx of S_8^{2-}, whereas based on relative extents of reduction events at the anode surface, its flux at the anode (i.e., consumption) changes sign (early on $S_{8(l)}$ reduction generates S_8^{2-} which is a positive flux and, later on, reduces to S_4^{2-} and in turn causes a negative flux; Figure 1f). Note that at each cross section, local charge neutrality is ensured. Given the presence of multiple anions, at each location, one has nontrivial profiles for ionic concentrations. Such speciation gives rise to a disparity between the influx of sulfur species at the cathode−separator interface (ions coming from the cathode to separator) and subsequent consumption of respective ions at the anode−separator interface (related to chemical redox reactions). This mismatch leads to accumulation (or attenuation) of a specific ion in the separator. In other words, even if ions enter the separator from the cathode side, the portion that gets reacted at the anode surface relies on the transport effectiveness of the electrolyte through the separator pore network. Figure 1f sketches these two fluxes for the species of interest: $S_{8(l)}$, S_8^{2-}, and S_4^{2-}. At the start of discharge, dissolved sulfur has the same concentration throughout the electrolyte phase (equal to solubility). It starts being consumed (chemically) at the anode surface (hence negative flux—solid line) and (electrochemically) inside the cathode (i.e., diffuses from the separator to cathode and the positive influx—dashed line). Later on, once longer-chain polysulfides are generated at the cathode, they start being transported to the anode. Simultaneously, first, the longer-chain polysulfides are generated at the anode and subsequently consumed, which explains the change in sign of the consumption flux. This generation of intermediate-chain polysulfides at the anode gives rise to an outflux of the ions to cathode until their concentration due to electrochemical reduction at the cathode reverses this gradient and makes it an influx (this inversion is apparent in Figure 1f). The direction of this flux might further change if electrodeposition of Li_2S in the cathode is sufficient to reduce the intermediate polysulfide concentration at the cathode and in effect causes an inflow. In nutshell, polysulfides do not always move from the cathode to anode, even when the net ionic current flows from the anode to cathode because the species flux is composed of diffusion and migration contributions. Their fluxes depend on local concentration gradients which in turn are closely related to the rates of chemical reactions at the anode and electrochemical reactions at the cathode. Diffusion directly involves concentration gradients, whereas migration indirectly depends on local concentration via transference numbers (transference numbers quantify the portion of the ionic current being carried by the respective ion).

■ IMPORTANCE OF REACTION TIMESCALES

Given the correspondence among the anode and cathode reactions, their rate constants are interrelated. The severity of anode surface reactions can be characterized by chemical overpotential, η_{ch} (expressions (S29−S31)). As chemical overpotential approaches zero, the anode surface reactions are more spontaneous (thermodynamically favored) and in turn faster. These chemical reactions take place even under open-circuit conditions (i.e., no passage of current across the cell). After assembly, impregnated solid sulfur dissolves in the electrolyte (finite solubility) and in turn diffuses to the anode and experiences chemical reduction events. This effectively

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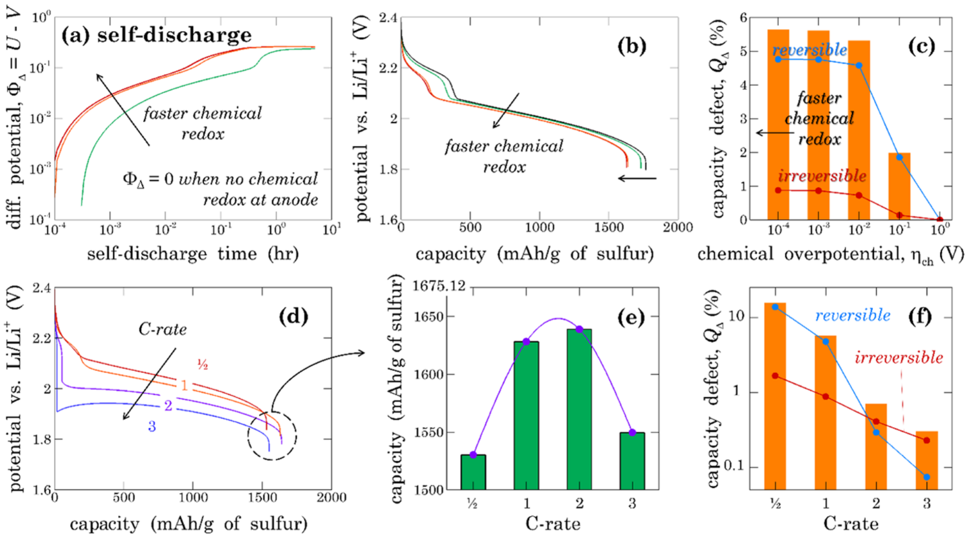

Figure 2. Effect of reaction time scales: (a) anode surface reactions gradually reduce sulfur and in turn limit the cell capacity; the severity of redox reactions is apparent from difference potential evolution during self-discharge; (b,c) at the constant C-rate, faster chemical redox means more capacity defect and shortened upper plateau; reversible and irreversible contributions also grow monotonically; (d−f) as the C-rate increases, the timescale of electrochemical reactions decreases which reduce the capacity defect, but cathode and electrolyte limitations become dominant causing a quadratic capacity trend.

increases the net charge per sulfur atom and equivalently the cell voltage and capacity diminish. This amounts to the self-discharge in LiS and can be quantitatively related to drop in open-circuit potential (referred to as the difference potential) $\Phi_\Delta = U - V$, where U is the open-circuit potential of the pristine cell and $V = V(t)$ (hence $\Phi_\Delta = \Phi_\Delta(t)$), identifies the evolution of the electrochemical state given the chemical redox reactions at the anode surface. The capacity defect and difference potential represent two different aspects of these redox interactions at the anode. Under self-discharge, all of the capacity loss is the capacity defect (with different time scales), and difference potential characterizes the severity of these unwanted reactions. On the other hand, during the electrochemical operation, multiple interactions take place simultaneously, and Q_Δ is a direct indicator. As the rate of these reactions is increased (i.e., $\eta_{ch} \to 0$), charge accumulation on sulfur is faster and in turn, the difference potential, $\Phi_\Delta(t)$, increases rapidly (Figure 2a). In other words, Φ_Δ measurements can shed light on the chemical reactivity of the anode. Figure 2b correlates the severity of this chemical redox reaction with changes in the electrochemical response. As chemical redox becomes severe, the capacity defect becomes more prominent. Interestingly, the width of the upper voltage plateau also reduces (1 C operation) which effectively relates to the fraction of dissolved sulfur participating in electrochemical reduction at the cathode. Corresponding capacity defects are sketched in Figure 2c along with their reversible and irreversible components. The capacity defect increases monotonically with faster anode reactions.

In contrast, Figure 2d−f explores the role of cathode electrochemical reactions (at constant polysulfide reactivity). As the C-rate is increased, the time scale for the electrochemical progression becomes smaller (chemical reaction time scales are rather intrinsically fixed), and one would expect more electrochemical reactions than chemical and subsequently smaller capacity defect (Figure 2e). As electrochemical reactions become faster, sulfur species experience faster electrochemical reduction. In fact, going from a low to a high C-rate, the cathode reaction order[29] gradually changes from successive to simultaneous as potential drops below the OCPs (open circuit potentials) for all reactions. This effectively alters the composition of the species influx from the cathode to the anode. At higher operating rates, the species influx contains a greater amount of reduced sulfur which results in a smaller quantity of chemical interactions at the anode surface (Figure 2f). There is an additional contribution here. Going toward higher rates, concentration gradients become sharper, and in turn, the disparity between the influx and consumption becomes starker (Figure S1) which amounts to the smaller amount of chemical redox at higher rates. This accounts for the increased capacity going from low to moderate rates of operation as plotted in Figure 2e, whereas at larger rates, physicochemical changes are more drastic at the cathode and lead to inferior performance.[29] These manifest as a nonlinear trend in cell capacity. Figure 2f exhibits an interesting trend: relative importance of reversible and irreversible defects qualitatively changes going to higher rates. With increasing rates, the amount of intermediate polysulfide increases in the incoming species flux. This makes the reduction of the intermediate polysulfide and irreversible formation of Li_2S at the anode the dominant anode reaction and in turn results in a higher irreversible capacity defect. On the other hand, at lower rates, higher order sulfur species do not electrochemically react fast enough, transport toward the anode and participate in reversible chemical reduction. Given the higher reactivity of these higher order sulfur species, contributions from the reversible capacity defect are greater when sulfur molecules reaching the anode are in the low enough reduction state.

MULTIMODAL INTERACTIONS AND COMPETING LIMITATIONS

The sulfur-to-electrolyte ratio (S/E) and cathode pristine porosity jointly identify the evolution of the cathode microstructure[29] and electrolyte transport. Both these phenomena affect the spatial distribution of the ionic species

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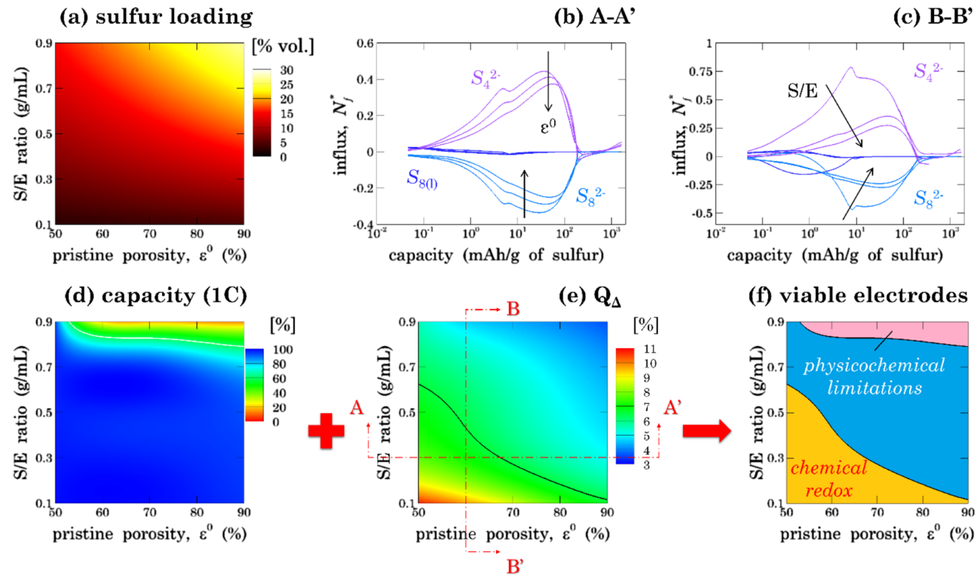

Figure 3. Cathode specifications dictate cell limitations: (a) volumetric sulfur loading as a function of S/E ratio and pristine porosity, (b) electrochemical capacity at 1 C and (e) capacity defect maps help identify (f) viable electrodes that have marginal limitations from various interactions, and (d) sulfur influx from the cathode to anode as cathode porosity is increased and (c) as S/E ratio is increased.

and consequently the anode surface reactions. Corresponding results are presented in Figure 3. Note that S/E and ε^0 together fix the volumetric sulfur loading

$$\varepsilon_S^0 = \frac{\varepsilon^0}{\left(1 + \frac{\rho_S}{S/E}\right)} \tag{6}$$

As mentioned earlier, the ionic current is proportional to the sulfur loading (S27 and S28), and consequently, the migration flux increases with increase in either S/E or ε^0 (Figure 3a). Curiously, the capacity defect decreases with either of these modifications (Figure 3e) because of diffusive transport and faster electrochemical reactions at the cathode. As sulfur loading is increased, operating current increases (at constant C-rate), which results in stronger concentration gradients and in turn diffusive transport is also important. Additionally, the electrochemical reactions become faster, and the sulfur species have a greater tendency to get reacted at the cathode rather than to transport to the anode. These two events cooperate with each other and effectively reduce the sulfur transport to the anode. These behavioral transitions are confirmed by the evolution of the species influx with pristine porosity (Figure 3b) and the sulfur-to-electrolyte ratio (Figure 3c). Thus, the capacity defect diminishes at higher ε^0 and/or S/E ratio. Figure S2 presents a comparison of the consumption and influx for the different conditions along cuts AA′ and BB′ in Figure 3e. The study of Figure S2 reveals that higher porosity (at constant S/E) reduces the influx and equivalently the consumption of each of these sulfur molecules. A similar trend is observed when S/E is increased at a constant pristine porosity (Figure S2). Observe that the small S/E ratio and high ε^0 both represent more electrolytes, yet their effect on the cell response is very different. Smaller S/E leads to more chemical reactions at the anode surface as the sulfur species are transported to the anode more efficiently, whereas higher ε^0 reduces the reaction rate as a more electrolyte volume is available at the cathode to retain more polysulfides.

At a higher S/E ratio (i.e., lean electrolyte conditions), cell capacity drops significantly in response to severe physicochemical changes taking place at the cathode and electrolyte, which comprises surface passivation, pore blockage, and concentrated solution effects (Figure 3d). Maps of the capacity (Figure 3d) and capacity defect (Figure 3e) paint a comprehensive picture of limitations in LiS. Figure 3f reveals that smaller S/E and electrode porosity result in an enhanced contribution from chemical redox at the anode. On the other hand, with higher sulfur content, a major source of limitation is a physical and chemical progression of the cathode and electrolyte. This identifies an intermediate range of cathode specifications where the limitations from each anode, electrolyte, and cathode are marginal.

■ COMPLEXATIONS ASSOCIATED WITH THE POLYSULFIDE SHUTTLE

Electrolyte-based sulfur species travel to the anode and undergoes chemical reduction, given the high reactivity of Li metal. The deleterious effects are caused by the reduction at the anode, whereas the reactant flux is controlled by a multitude of factors such as cathode evolution, electrolyte transport dynamics, and the rate of anode-centric chemical reactions. This transport of species, that is, shuttling, in itself is not detrimental. For instance, during self-discharge, there is no net current, that is, no driving force for polysulfides to travel from the cathode to anode. However, due to the reaction of sulfur species at the anode, a net flux exists. Thus, it is a reaction-limited extreme of the polysulfide shuttle effect. On the other hand, during electrochemical reduction of sulfur at the cathode (i.e., discharge), sulfur species travel to the anode in response to concentration and potential gradients. Potential gradients drive the transport from the cathode to anode, whereas concentrations evolve nonmonotonically and can give rise to fluxes in either direction. This is a shuttle-limited scenario where the extent of chemical reduction depends on the amount of sulfur species coming from the cathode to anode. There are various aspects of this extreme. If the electrochemical reactions are carried out faster, the flux reaching anode is in a more reduced state and effectively the severity of anode chemical reactions decreases. If the

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electrolyte is highly concentrated (i.e., high sulfur to electrolyte ratio), its transport effectiveness diminishes and results in smaller polysulfide flux and in turn less detrimental chemical reduction at the anode. In nutshell, the root cause of issues associated with the polysulfide shuttle effect is anode's ability to reduce incoming sulfur species and not the species flux (i.e., shuttle). The article title points to this fundamental distinction.

■ CONCLUSIONS

In conclusion, the polysulfide shuttle effect is much more intricate than the phenomenological arguments about its deleterious contributions. The physicochemical investigation reveals that it is composed of two distinct effects: transport of sulfur species to the anode and subsequent chemical reduction at the anode surface. These redox reactions give rise to reversible and irreversible capacity defects. The transport itself is composed of diffusive and migration contributions. Even if the migration flux is always from the cathode to anode, speciation and reaction rates dictate the diffusive fluxes and in turn net transport. Given the multitude of factors affecting transport, the extent of the polysulfide shuttle effect demonstrates a nonlinear dependence on operating conditions. Under lean electrolyte conditions (high S/E ratio), hindered species transport limits the negative contribution of the polysulfide shuttle effect and rather cathode and electrolyte phase interactions become performance limiting.

■ ASSOCIATED CONTENT

Ⓢ Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jpcc.8b06077.

> The mathematical details of the physicochemical analysis are presented in the Supporting Information. S1: Chemical interactions in the Li−sulfur cell, S2: Electrochemical description for the Li−sulfur cell, S3: Sulfur balance sheet, and S4: Physicochemical interactions in the separator (PDF)

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Notes

The authors declare no competing financial interest.

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